6. **(a)** Explain what is meant by photoelectric effect.

.....

.....

[1]

.....

(b) In a photoelectric experiment, a parallel beam of monochromatic radiation is incident normally upon a metal surface of area 1.0×10^{-4} m^2 in a vacuum tube. The metal has a work function of 2.06 eV. The photocurrent I against p.d. V graph is shown in Fig. 6.1 below.

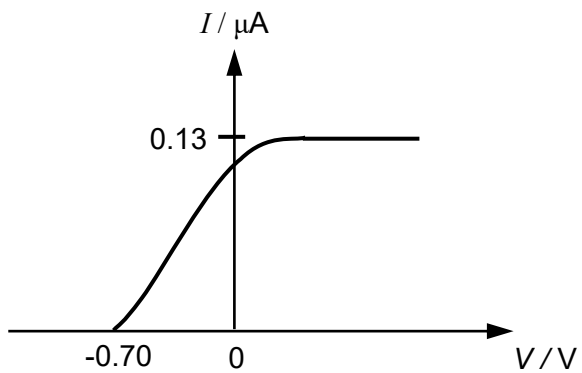

Fig. 6.1

(i) Calculate the maximum kinetic energy of the photoelectrons emitted.

maximum kinetic energy = ... J [1]

(ii) Hence, determine the frequency of radiation incident on the metal surface.

frequency = ... Hz [2]

(iii) If one photoelectron is emitted for every 8000 photons incident on the metal, calculate the intensity of the radiation incident on the metal surface.

intensity = ... W m^{-2} [3]

(iv) Sketch on Fig. 6.1 the graph you expect to obtain if the intensity of the radiation is halved. [1]

(v) Suggest a reason for the part of the graph in Fig. 6.1 for negative values of *V*.

.....

.....

..... [2]